

Pure Mathematics · Week 3

8 October 2026 · Original practice · Student copy · No solutions

Try independently, compare with a partner, then explain why your method works. Write complete reasons. These are practice activities, not hand-in questions.

Activity 1

Let $S = \{m \in \mathbb{Z} \geq 0 : m > 17/4\}$. List four elements and find its least element. Explain why well-ordering applies. Contrast this with $T = \{x \in \mathbb{R} : x > 17/4\}$: does T have a least element?

Activity 2

Find integers q, r such that (a) $-29 = 6q + r$ with $0 \leq r < 6$; (b) $29 = (-6)q + r$ with $0 \leq r < 6$. In (a), a student writes $q = -4, r = -5$. Is the equation correct? Is this the required division-algorithm answer? If the remainder restriction is removed, exhibit two answers.

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Suppose $-29 = 6q + r = 6q' + r'$, where q, q', r, r' are integers and $0 \leq r, r' < 6$. Prove the two representations must agree. Your proof must use both the equality and the remainder bounds.

Activity 4

(a) Prove that if 4 divides u and 4 divides v , then 4 divides $3u - 2v$. (b) A student claims that $8 \mid 3u - 2v$ follows too. Decide by proof or counterexample. (c) List the positive common divisors of 18 and 30 and identify their gcd.

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You are given $6 = 2 \cdot 18 - 30$. Use it to prove that every common divisor of 18 and 30 divides 6. Then explain why it certifies $\gcd(18, 30) = 6$. Finally, can integers x, y satisfy $18x + 30y = 9$?

Activity 6

Without notes: (1) Find q, r for $22 = (-5)q + r$ with $0 \leq r < 5$. (2) State the two facts needed to certify that a positive integer g is $\gcd(a, b)$. (3) Name one hypothesis of well-ordering that you must check.

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8 October 2026 · Original practice · Student copy · No solutions

Try independently, compare with a partner, then explain why your method works. Write complete reasons. These are practice activities, not hand-in questions.

Activity 1

Let $S = \{m \in \mathbb{Z} \geq 0 : m > 17/4\}$. List four elements and find its least element. Explain why well-ordering applies. Contrast this with $T = \{x \in \mathbb{R} : x > 17/4\}$: does T have a least element?

Activity 2

Find integers q, r such that (a) $-29 = 6q + r$ with $0 \leq r < 6$; (b) $29 = (-6)q + r$ with $0 \leq r < 6$. In (a), a student writes $q = -4, r = -5$. Is the equation correct? Is this the required division-algorithm answer? If the remainder restriction is removed, exhibit two answers.

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Activity 3

Suppose $-29 = 6q + r = 6q' + r'$, where q, q', r, r' are integers and $0 \leq r, r' < 6$. Prove the two representations must agree. Your proof must use both the equality and the remainder bounds.

Activity 4

(a) Prove that if 4 divides u and 4 divides v , then 4 divides $3u - 2v$. (b) A student claims that $8 \mid 3u - 2v$ follows too. Decide by proof or counterexample. (c) List the positive common divisors of 18 and 30 and identify their gcd.

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Activity 5

You are given $6 = 2 \cdot 18 - 30$. Use it to prove that every common divisor of 18 and 30 divides 6. Then explain why it certifies $\gcd(18, 30) = 6$. Finally, can integers x, y satisfy $18x + 30y = 9$?

Activity 6

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